



Modelling Financial Markets

Chapter 16: Digital Assets and DeFi



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Acronyms used in this material (1/2)

- AI: Artificial Intelligence
- AIC: Akaike Information Criterion
- AR: AutoRegressive (model); in event studies: Abnormal Return
- BET: Bucharest Exchange Trading (index)
- BNB: BNB (native token of BNB Chain, formerly Binance Coin)
- BSC: BNB Smart Chain
- CAPM: Capital Asset Pricing Model
- CI: Confidence Interval
- CMKT: Crypto Market factor (value-weighted market excess return)
- CMOM: Crypto Momentum factor (past winners minus losers)
- COVID-19: COronaVirus Disease 2019
- CRIX: CRypto IndeX
- CSMB: Crypto Size factor (small minus big coins)
- DAI: Dai (stablecoin of the Maker/Sky protocol)
- DTB4WK: 4-Week Treasury Bill Secondary Market Rate (FRED series code)
- ECRIX: Exact CRIX (number of constituents chosen in steps of one)
- EFCRIX: Exact Full CRIX (global minimum of the AIC)
- ES: Expected Shortfall
- ETF: Exchange-Traded Fund
- ETHA: iShares Ethereum Trust ETF (ticker)
- FDIC: Federal Deposit Insurance Corporation
- FM: Fama–MacBeth (regressions)
- FRED: Federal Reserve Economic Data
- FWER: Family-Wise Error Rate
- GMM: Generalized Method of Moments
- GRS: Gibbons–Ross–Shanken (test)

Acronyms used in this material (2/2)

- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IA: Internet Appendix (of the cited paper)
- ▣ LP: Liquidity Provider
- ▣ LUNA: Luna (native token of the Terra blockchain)
- ▣ LVR: Loss-Versus-Rebalancing
- ▣ MDD: Maximum DrawDown
- ▣ OLS: Ordinary Least Squares
- ▣ PC1: first Principal Component
- ▣ PCA: Principal Component Analysis
- ▣ SE: Standard Error
- ▣ SVB: Silicon Valley Bank
- ▣ TAR: Threshold AutoRegression
- ▣ TE: Tracking Error
- ▣ TMI: Total Market Index (all coins, CRIX methodology)
- ▣ TVL: Total Value Locked
- ▣ US: United States
- ▣ USDC: USD Coin (stablecoin issued by Circle)
- ▣ UST: TerraUSD (algorithmic stablecoin of the Terra blockchain)
- ▣ XRP: XRP (the native token of the XRP Ledger)

Chapter Route and Outcomes

Route of this chapter

- ▣ Part I: the crypto market, stylised facts of returns, crypto indices (CRIX), stablecoins
- ▣ Part II: DeFi and automated market makers, spot ETFs and crypto equities, cryptos as an asset class, a case study, one pass of the AI-assisted research loop
- ▣ Technical detail sits in the [Appendix](#) at the end

After this chapter you can

- ▣ measure a 7-day market on the right calendar and attach intervals to its tail index, volatility and correlations
- ▣ build a market-value index with a divisor and model a stablecoin peg as a threshold AR
- ▣ price a swap in a constant-product pool and explain why providing liquidity is selling volatility
- ▣ judge ETF tracking, crypto betas and asset-class claims with tests that respect closing times and unknown break dates

Blockchains, Coins and Tokens



Bust of Satoshi Nakamoto, Graphisoft Park,
Budapest

Photo: Fekist (2021); CC BY-SA 4.0; Wikimedia
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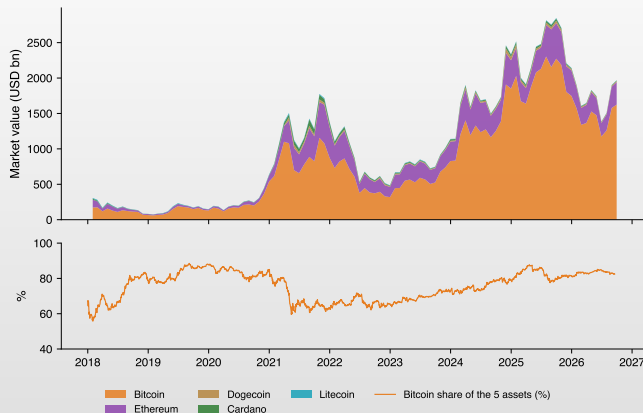
- **Blockchain Nakamoto (2008)**: a shared ledger of blocks, each carrying the hash of the previous one; secured by proof of work (Bitcoin) or proof of stake (Ethereum since 15 September 2022, ethereum.org: The Merge)
- Bitcoin issuance halves over time up to 21 million coins [Bitcoin.org](https://bitcoin.org) FAQ; 20.09 million existed on 18 Sep 2026
- **Coins** are native assets of a blockchain; **tokens** (stablecoins, governance tokens, tokenised funds) are issued by **smart contracts** Buterin (2014); DeFi (decentralised finance) is built from such contracts Schär (2021)
- Surveys: technology Lin, Pele & Ren (2025); economics Hårdle, Harvey & Reule (2020)

Data Used in This Chapter

Series	Content	Period
Crypto prices	Bitcoin, Ethereum, Solana, Dogecoin, Cardano, Litecoin; daily, 7 days a week	2018–2026
Stablecoins	USDT, USDC, DAI prices; supply by coin (De-fiLlama)	2019–2026
Coin supply	Coins in circulation, daily (Coin Metrics Community Data)	2018–2026
DeFi	Total value locked, all chains and by chain (De-fiLlama)	2020–2026
ETFs and shares	spot ETFs on Bitcoin (IBIT) and Ether (ETHA), Coinbase, Strategy; S&P 500, Nasdaq 100, gold	2018–2026
Asset universe	26 assets: cryptos, equity indices, currencies, commodities, bonds	2019–2026

- Daily closes up to 18 September 2026; crypto on its own 7-day calendar; analyses that combine assets use common days (prices aligned on common trading days first, then returns)

Five Coins, One Dominant Asset



- Market value $M_{i,t} = P_{i,t} \times Q_{i,t}$ (price in USD times coins in circulation), month ends; Bitcoin: 1.63 trillion USD on 18 Sep 2026
- The five coins: 1970 billion USD today, peak 3135 billion (6 Oct 2025); the value moves with prices, supply changes slowly
- Bitcoin's share: 82.5% today, between 55.8% (1 Feb 2018) and 88.5% (5 Sep 2019)
- Herfindahl–Hirschman index** $HHI = \sum_i w_i^2$ (w_i : value shares) = 0.71: an effective number of coins $1/HHI = 1.42$
- XRP and Chainlink are left out: their reported supply includes issuer-held tokens (free float matters)

Measures and the Tail Index

- ▣ Daily log return in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$, each series on its own calendar; $\sigma_a = s\sqrt{m}$, m = observations per year (about 365 for crypto, 252 for shares)
- ▣ Skewness S ; MDD = $\min_t (P_t / \max_{u \leq t} P_u - 1)$; volatility clustering $\hat{\rho}_1(|r|)$ (Chapter 1); historical daily $\text{VaR}_{1\%} = -q_{1\%}(r)$ and $\text{ES}_{2.5\%} = -\overline{r \mid r \leq q_{2.5\%}(r)}$ (Chapter 7)
- ▣ Hill estimator [Hill \(1975\)](#) from the k largest losses $L_{(1)} \geq \dots \geq L_{(k+1)}$, $k = 5\%$ of the losses:

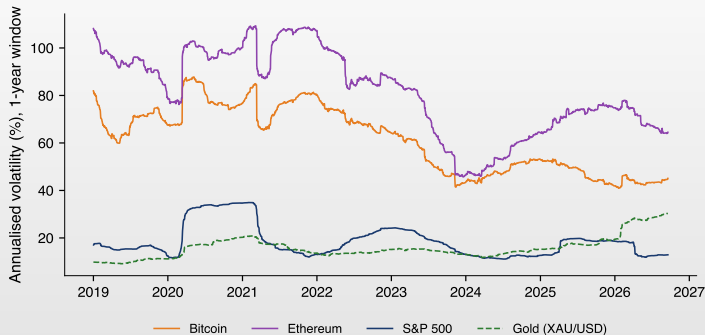
$$\hat{\alpha}_H = \left(\frac{1}{k} \sum_{j=1}^k \ln(L_{(j)} / L_{(k+1)}) \right)^{-1}$$
 - ▶ $P(L > x) \sim Cx^{-\alpha}$: smaller α = heavier tail; $\text{SE} = \hat{\alpha}_H / \sqrt{k}$ for independent and identically distributed (i.i.d.) data (conditions and dependence in the [Appendix](#))
- ▣ All six i.i.d. 95% CIs lie below 3.7: $\alpha < 4$, hence $E|r|^4 = \infty$, is plausible but not proven
 - ▶ If $\alpha < 4$, the sample excess kurtosis $K = \overline{(r - \bar{r})^4} / s^4 - 3$ estimates nothing: report $\hat{\alpha}_H$ with its CI instead
- ▣ Sample 2018–2026 (Solana from April 2020); comparison assets: S&P 500 and gold

Crypto vs Shares and Gold, 2018–2026

	n	σ_a (%)	S	MDD (%)	$\hat{\rho}_1(r)$	$\hat{\alpha}_H$	95% CI	VaR _{1%}	ES _{2.5%}
Bitcoin	3 182	64	−0.95	−82	0.17	2.73	[2.12, 3.33]	10.2	10.3
Ethereum	3 182	84	−0.82	−94	0.16	2.67	[2.08, 3.26]	13.5	13.7
Solana	2 351	117	−0.22	−96	0.28	2.85	[2.12, 3.58]	14.5	17.1
Dogecoin	3 182	124	4.88	−92	0.37	2.83	[2.22, 3.45]	13.8	16.4
S&P 500	2 189	19	−0.65	−34	0.36	2.87	[2.07, 3.67]	3.4	3.8
Gold	2 269	17	−0.55	−28	0.13	2.75	[1.99, 3.51]	3.0	3.2

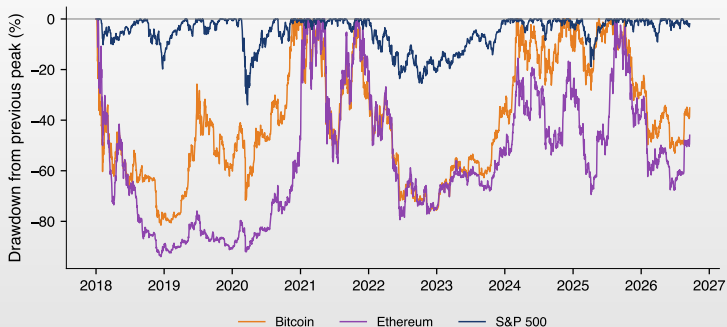
- Daily log returns in %; VaR and ES are losses of the daily log return; $\hat{\alpha}_H$ from the largest 5% of losses, CI $\hat{\alpha}_H \pm 1.96 \hat{\alpha}_H / \sqrt{k}$
- Bitcoin: 3.3 times the S&P 500's volatility; daily VaR 1% 10.2% in log terms, a loss of 9.7% of the position; worst day 12 Mar 2020 (COVID-19 crash)
- Drawdowns of 82–96% for the coins, 34% for the S&P 500; Hill indices 2.67–2.87 for all six assets: crypto is more volatile, but the data give no strong evidence of different tail indices (tests in the [Appendix](#))

Volatility Over Time



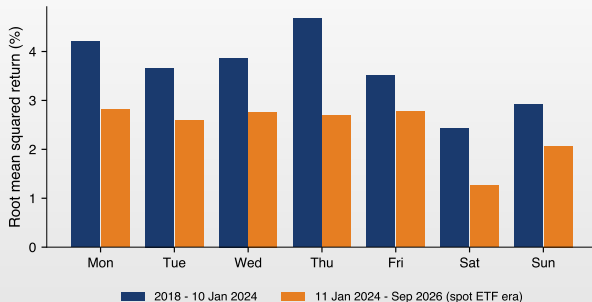
- Annualised standard deviation of daily log returns over a moving one-year window
- Bitcoin fell from 88% to 45% today, Ethereum from 109% to 64%: consistent with a maturing market, but the chart does not identify the cause
- Bitcoin is still 3.5 times as volatile as the S&P 500 (13%); gold reached 30% in 2026, a record for the sample

Drawdowns: How Deep and How Long



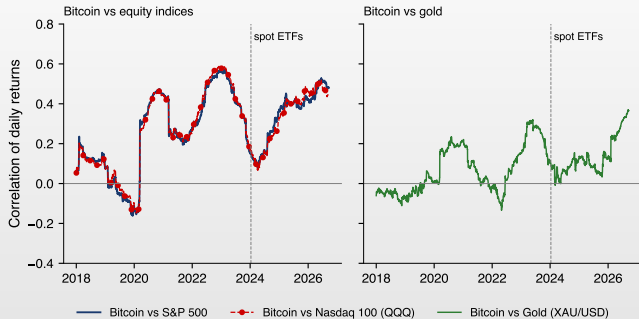
- Fall from the previous maximum, in %: Bitcoin was more than 50% below its peak on 36% of the days since 2018, Ethereum on 61%; the S&P 500 at most 34% (March 2020)
- Today: Bitcoin 35%, Ethereum 46% below their peaks; long “crypto winters” (2018–2020, 2022–2023) follow each boom (bubbles: Chapter 17)

A 7-Day Market with a Weekend Lull



- Root mean squared daily Bitcoin return by day of the week; weekend variance / weekday variance: 0.45 before and 0.39 after 11 January 2024
- Bitcoin trades on Saturdays, but macro news, US equity trading and ETF flows do not; all days calmer after the ETF launch (weekday root mean square 4.01% → 2.73%)
- Annualise with 365 days, but do not treat all days as identical; did the ratio change? Seminar 16, B1

Is Bitcoin Moving with Shares?



- Correlation of daily log returns over 250 common days (prices aligned on common trading days first, then returns); Bitcoin vs S&P 500: 0.02 (2017–2019), 0.39 (2020–2023), 0.38 since the ETFs (674 days)
- Nasdaq 100 like the S&P 500 (0.38); gold -0.02 , 0.12 , 0.22 : a smaller diversification benefit than before 2020 [Bouri et al. \(2017\)](#), [Liu & Tsyvinski \(2021\)](#)
- The chart suggests a break in early 2020, not in January 2024: tested on the next two slides

Did the Bitcoin–Equity Link Break in 2020? A Formal Test

- Model on 2 692 common days since January 2016: $r_t^{\text{BTC}} = a + b r_t^{\text{SPX}} + u_t$, with (a, b) allowed to shift at an unknown date τ
 - ▶ sup-Wald [Andrews \(1993\)](#): $\sup_{\tau \in [0.15T, 0.85T]} W(\tau)$, $W(\tau)$ the Wald statistic for no change, with Newey–West covariances (Bartlett, 8 lags) in each regime
 - ▶ Under H_0 the sup is not χ_2^2 : its limit is $\sup_r \|B(r) - rB(1)\|^2 / [r(1-r)]$ (B a Brownian motion), simulated here: 5% critical value 11.55
- Result: $\sup W = 20.21$ ($p = 0.0008$); least-squares break date 9 Mar 2020 ($W = 18.81$); at the ETF launch $W = 2.31$
 - ▶ 95% CI for the date [Bai \(1997\)](#): 27 Nov 2018 – 15 Jun 2021; it excludes January 2024
 - ▶ Beta to the S&P 500: 0.07 before, 1.22 after; correlation 0.01 and 0.40
- Several breaks at unknown dates: the sequential tests and CIs of [Bai & Perron \(1998\)](#)

Correlation or Volatility? The Forbes–Rigobon Adjustment

- Pearson correlation rises mechanically when the volatility of the driving market rises, with the same underlying link [Forbes & Rigobon \(2002\)](#)
 - ▶ Adjusted correlation: $\rho^* = \rho / \sqrt{1 + \delta(1 - \rho^2)}$, $\delta = \sigma_{\text{SPX,high}}^2 / \sigma_{\text{SPX,low}}^2 - 1$, with 2017–2019 as the calm base period
- 2020–2023: S&P 500 volatility 23.1% vs 12.8% ($\delta = 2.24$); $\hat{\rho} = 0.39$ becomes $\rho^* = 0.23$
 - ▶ Since the ETFs: $\hat{\rho} = 0.38$, $\rho^* = 0.33$ ($\delta = 0.42$)
- Change against 2017–2019 ($\hat{\rho} = 0.02$): $\rho_{2020-23}^* - \hat{\rho}_{2017-19} = 0.21$, 95% block-bootstrap CI [0.09, 0.34]
 - ▶ Under the adjustment's assumptions, 44% of the raw rise is attributed to volatility; the adjusted increase remains positive: stronger co-movement, not an identified mechanism
- Caveat: the adjustment assumes no endogeneity and no omitted common shocks, which a pandemic hardly satisfies

A Crypto Index and Its Divisor



Humboldt-Universität zu Berlin, where CRIX was developed

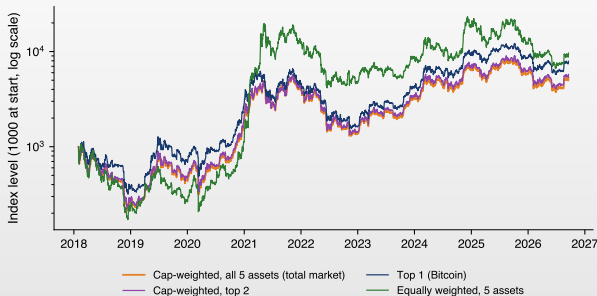
Photo: Diego Delso (2016); CC BY-SA 4.0;
Wikimedia Commons

- ▣ **CRIX Trimborn & Härdle (2018)**: a market-value weighted benchmark of the crypto market, published at thecrix.de; review of crypto indices: Găman, Pele, Härdle & Będowska-Sójka (2026)
- ▣ Level on day t of reallocation period τ : $I_t = \sum_{i=1}^k P_{i,t} Q_{i,\tau} / D_\tau$
 - ▶ $Q_{i,\tau}$: coins in circulation, fixed within the period; D_τ : the **divisor**
 - ▶ Index return = weighted average of the constituents' returns, weights = value shares at the start of the period; a Laspeyres-type index (quantities fixed within the period)
- ▣ At a reallocation (new Q , new members) the divisor keeps the level: $D_{\tau+1} = \sum_i P_{i,t} Q_{i,\tau+1} / I_t$
 - ▶ Here: monthly reallocation, weights from the previous month end; worked examples: Seminar 16, A1–A2

How Many Constituents? The CRIX Rule

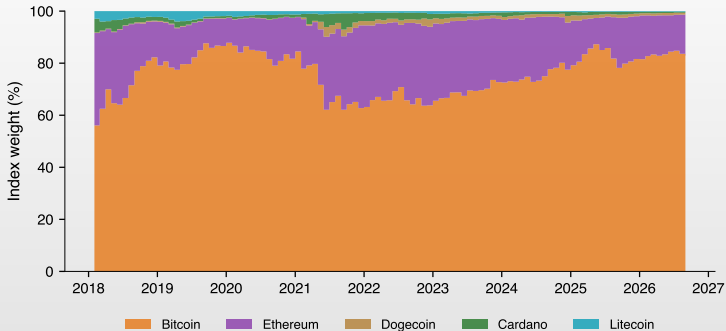
- Candidate index $\text{CRIX}(k, \beta)$: the k_1 largest coins at weight 1, the next $s = k - k_1$ with free weights β , fitted to track the total market index TMI (all coins) [Trimborn & Härdle \(2018\)](#)
 - ▶ Tracking residual $\hat{\varepsilon}_t(k, \beta) = \Delta \ln \text{TMI}_t - \Delta \ln \text{CRIX}_t(k, \beta)$
- $\text{AIC}(k) = -2 \log L\{\hat{\varepsilon}(k, \hat{\beta})\} + 2s$: more coins track better but cost parameters
 - ▶ k reviewed quarterly on the last three months of data; members reallocated monthly by market value
 - ▶ Two stopping rules, ECRIX (first rise of the AIC) and EFCRIX (global minimum); likelihood details in the [Appendix](#)
- Data limitation: circulating supply is public for five of our coins only (Bitcoin, Ethereum, Dogecoin, Cardano, Litecoin), so $k \leq 4$ and TMI has five members, not hundreds

Crypto Indices Since 2018



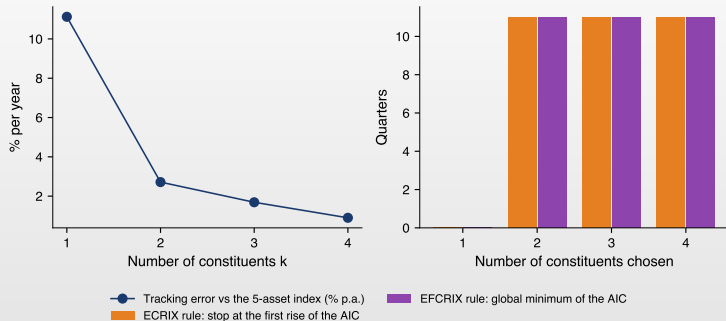
- Log scale; total-market index 21.4% a year over 8.6 years $((I_T/I_0)^{1/\text{years}} - 1)$, volatility 65.7%; Bitcoin alone did better: the other coins lost weight and value
- Equal weights: 30.0% a year at 82.5% volatility, driven by 2021 (+998% vs +113%): a few extreme months, not a stable premium
- An index is only as good as its universe: today's survivors are not the leaders of 2018 Będowska-Sójka, Wójcik & Pele (2026)

Index Weights Over Time



- Value weights at each monthly reallocation, %: Bitcoin 75.7% on average, never below 56.1%, today 82.9%; Ethereum 15.8%; the other three together below 2%
- Altcoins (coins other than Bitcoin) gain weight late in booms (2021) and lose it in busts: a value-weighted crypto index is mostly Bitcoin

Tracking Error and the CRIX Rule



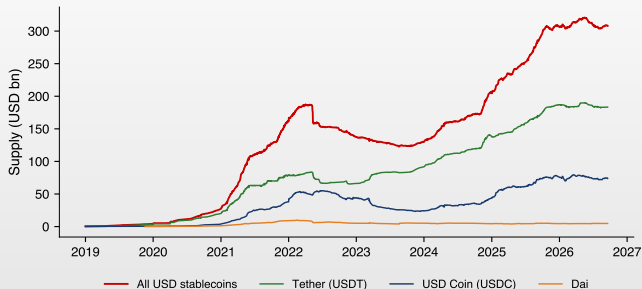
- Left: tracking error of the top- k index (weights 1) against the five-coin index: 11.1% a year for Bitcoin alone, 2.7% for two coins, 1.7% for three, 0.9% for four
- Right: the rule picks 2 coins in 11, 3 in 11 and 4 in 11 of 33 quarters (2 in the last); ECRIX and EFCRIX agree in every quarter

How a Stablecoin Holds 1 USD

Design	Backing	Examples (USD bn, 18 Sep 2026)
Fiat-backed	Bank deposits and short US Treasury bills held by the issuer	Tether (USDT) 183.33, USD Coin (USDC) 73.92, PayPal USD (PYUSD) 2.81
Crypto-backed	Over-collateralised crypto deposits or hedged positions, managed by smart contracts	Sky Dollar (USDS) 6.49, Ethena USDe 4.74, Dai (DAI) 4.77
Algorithmic	No external collateral: supply expands and contracts against a second token	TerraUSD (collapsed in May 2022)

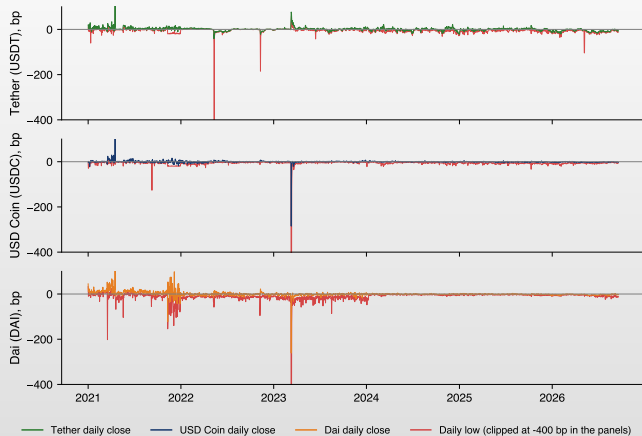
- Authorised traders **mint** (pay 1 USD, receive 1 coin) and **redeem** (return 1 coin, receive 1 USD) with the issuer: with costs c_r and c_m , arbitrage keeps the price in the band $1 - c_r \leq P \leq 1 + c_m$
 - Below the band: buy and redeem; above: mint and sell (evidence: [Lyons & Viswanath-Natraj \(2023\)](#); Tether issuance and Bitcoin prices: [Griffin & Shams \(2020\)](#))
- The band breaks when redemption stops or the backing is doubted; reserve and redemption rules: [MiCA, Regulation \(EU\) 2023/1114](#), [GENIUS Act](#), [Public Law 119-27](#)

The Rise of Stablecoins



- Supply in circulation ([DefiLlama](#)): from 4.2 billion USD in January 2020 to 308 billion on 18 Sep 2026, about 74 times: the cash leg of crypto trading and DeFi
- Tether 183 billion (60%), USD Coin 74 billion (24%); fiat-backed 283, crypto-backed 22.4, algorithmic 0.15 billion
- 2022–2023 dip: 187 billion (2 Apr 2022) to 123 billion (19 Aug 2023); TerraUSD vanished, USD Coin fell from 55 to 23 billion

Deviations from 1 USD



- Daily close and daily low minus 1 USD, in basis points ($1 \text{ bp} = 0.01\%$), 2021–2026; lows below -400 bp are cut at the panel edge
- Normal times: standard deviation of the close 7.3 bp (Tether), 7.5 (USD Coin), 10.8 (Dai); beyond 50 bp on less than 1% of the days
- Stress is rare but deep: USD Coin's low -1226 bp (11 Mar 2023), Tether's -515 bp (12 May 2022)
- A stablecoin is a claim with a short put on the reserves: nothing happens most days, a large loss is possible on a few

Arbitrage Bands Imply a Threshold AR

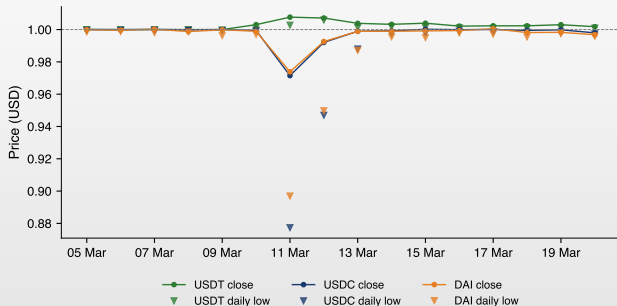
- Deviation from the peg in basis points (1 bp = 0.01%): $d_t = 10^4(P_t - 1)$, daily close
 - ▶ Inside the band $|d| \leq c$ arbitrage does not pay: weak pull; outside it does: fast reversion
- Threshold autoregression (TAR), the equilibrium form of [Balke & Fomby \(1997\)](#): $d_t = \phi_{\text{in}} d_{t-1} + \varepsilon_t$ if $|d_{t-1}| \leq c$, $d_t = \phi_{\text{out}} d_{t-1} + \varepsilon_t$ otherwise
 - ▶ $\hat{c}$ by least squares over a grid of $|d_{t-1}|$ values, keeping at least 15% of the days in each regime
 - ▶ $\hat{c}$ (in bp) is an empirical regime boundary: it matches the costs only under a symmetric band $c_r = c_m$ (USD per token), as $10^4 c_r$ bp, and other frictions also move it
- Test of linearity $H_0: \phi_{\text{in}} = \phi_{\text{out}}$: c is not identified under H_0 , so $\sup_c W(c)$ is not χ_1^2 (the Davies problem)
 - ▶ p -value from the fixed-regressor bootstrap of [Hansen \(1996\)](#) with heteroskedasticity: $d_t^* = \hat{\varepsilon}_t \eta_t$, $\eta_t \sim N(0, 1)$, regressors held fixed; $W(c)$ with White standard errors
- Sample 2021–2026: earlier quotes are thin and noisy

Threshold AR Estimates for the Three Stablecoins

	$\hat{c}$ (bp)	$\hat{\phi}_{\text{in}}$ (SE)	$\hat{\phi}_{\text{out}}$ (SE)	sup W	p bootstrap	p (χ_1^2)	$\hat{\phi}$ linear
Tether	8.6	0.86 (0.02)	0.69 (0.11)	7.4	0.084	0.006	0.73
USD Coin	2.9	0.59 (0.04)	0.28 (0.03)	32.5	0.001	< 0.001	0.29
Dai	6.0	0.72 (0.08)	0.32 (0.07)	13.8	0.005	< 0.001	0.33

- ▣ USD Coin and Dai: linearity rejected; reversion is much faster outside the band (half-life 0.5 vs 1.3 days for USD Coin)
 - ▶ Without 9–16 March 2023, USD Coin gives $p = 0.29$ (bootstrap) but 0.019 (χ_1^2): one stress week carries the evidence, and χ^2 over-rejects
- ▣ Tether: the widest band ($\hat{c} = 8.6$ bp) and slow reversion inside it, but linearity is not rejected at 5%: consistent with fewer direct redeemers [Ma, Zeng & Zhang \(2025\)](#)

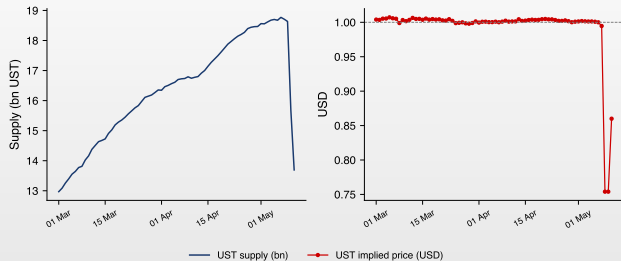
March 2023: a Bank Failure Hits USD Coin



- Daily close (lines) and daily low (triangles), 5–20 March 2023
- Friday 10 March: Silicon Valley Bank (SVB) is closed, the FDIC is appointed receiver [FDIC \(2023\)](#); USD Coin held part of its reserves there [Diop, Chevallier & Sanhaji \(2024\)](#)
- Weekend: redemptions could not settle, the band disappeared; USD Coin's low 0.8774 USD on Saturday 11 March (close 0.9715); Dai, crypto-backed, fell to 0.8970
- Tether rose to 1.0077 (high 1.0296): a flight to the other large stablecoin
- Sunday 12 March: all SVB depositors protected [Treasury, Federal Reserve & FDIC \(2023\)](#); the peg returned within two days: the event tested the backing, not the design

 [MFM_ch16_stablecoins](#)

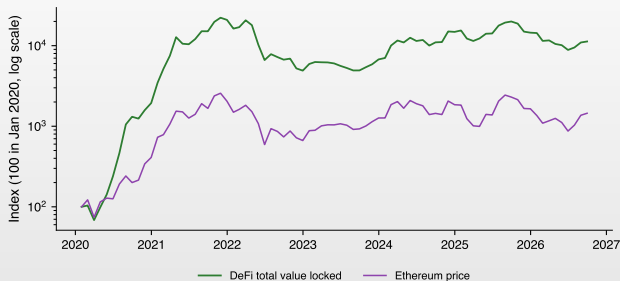
TerraUSD: an Algorithmic Stablecoin Collapses



- ▣ Supply and implied price (market value / supply), March–May 2022 ([DefiLlama](#))
- ▣ TerraUSD (UST) held its peg by exchange with the token LUNA, not with reserves; demand came from the high yields of the Anchor lending protocol [Liu, Makarov & Schoar \(2023\)](#)
- ▣ Supply 18.8 billion on 7 May 2022, 13.7 billion four days later (27% redeemed); implied price 0.754 USD on 10 May 2022
- ▣ Redemption minted more LUNA and pushed its price down: a death spiral that wiped out 50 billion USD in three days; sophisticated holders ran first and lost less [Liu, Makarov & Schoar \(2023\)](#)
- ▣ Without external collateral the peg rests on confidence alone: a run is self-fulfilling

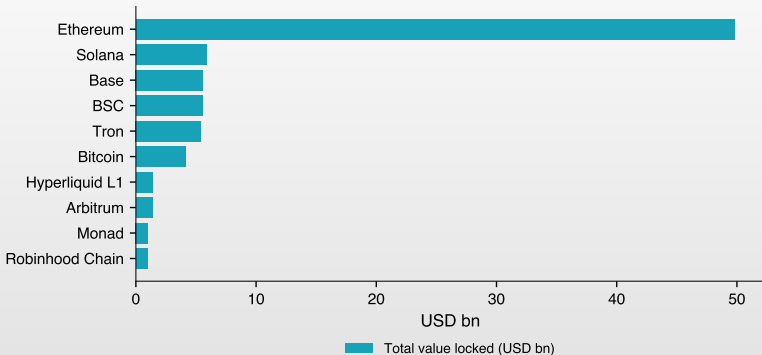
 [MFM_ch16_stablecoins](#)

Total Value Locked in DeFi



- Total value locked (TVL): the value of tokens deposited in DeFi protocols, all chains ([DefiLlama](#)), and the Ether price; month ends, 100 in January 2020, log scale
- TVL: 0.6 billion USD in January 2020, 177 billion at the peak (9 Nov 2021), 88 billion today (50% below the peak)
- Monthly $\Delta \ln \text{TVL}_t = a + b \Delta \ln P_t^{\text{ETH}} + u_t$,
Newey–West SE: $\hat{b} = 0.79$ (SE 0.07),
 $R^2 = 0.62$, $n = 80$
- TVL mixes token prices with net deposits: measure activity with fixed-price quantities, net flows or fees; DeFi risks: leverage, liquidity mismatch, interconnection [Aramonte, Huang & Schrimpf \(2021\)](#)

Where the Value Is Locked



- TVL by blockchain, ten largest, 18 September 2026 ([DefiLlama](#)): Ethereum 49.8 billion USD, 57% of 88 billion; Solana 5.9, Base 5.6, BSC (BNB Smart Chain) 5.6, Tron 5.4
- Base and Arbitrum are layer-2 networks that settle on Ethereum: cheaper transactions, the same security anchor

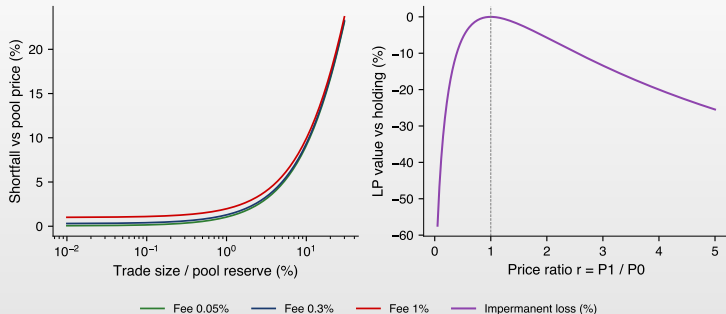
The Constant-Product Market Maker

- A pool holds reserves x of token X (e.g. Ether) and y of token Y (e.g. USD Coin); it accepts any trade that keeps $x \cdot y = k$ [Adams, Zinsmeister & Robinson \(2020\)](#)
 - ▶ Pool price (marginal): $p = y/x$ units of Y per X
- A trader sells Δx ; the pool keeps a fee f , so $\gamma \Delta x$ enters the invariant, $\gamma = 1 - f$
 - ▶ $(x + \gamma \Delta x)(y - \Delta y) = xy \Rightarrow \Delta y = \frac{y \gamma \Delta x}{x + \gamma \Delta x}$
- Execution price $\Delta y / \Delta x = p \frac{\gamma}{1 + \gamma \delta}$, with $\delta = \Delta x / x$ the trade size relative to the reserve
 - ▶ **Shortfall** against the pool price: $1 - \gamma / (1 + \gamma \delta)$: fee plus price impact
- Arbitrageurs keep p in line with other exchanges: the pool price follows the market [Angeris et al. \(2019\)](#)

Worked Example: a Swap in an Ether Pool

- Pool: $x = 2\,000$ Ether, $y = 5\,000\,000$ USD Coin, so $p = 2\,500$ and $k = 10\,000 \times 10^6$; fee $f = 0.3\%$
 - ▶ Sell $\Delta x = 20$ Ether ($\delta = 1\%$); after the fee, $\gamma \Delta x = 19.94$
- $\Delta y = \frac{5\,000\,000 \times 19.94}{2\,000 + 19.94} = 49\,358$ USD Coin, an execution price of 2467.9 per Ether
- Shortfall: $1 - 2467.9/2\,500 = 1.28\%$ (0.3% fee plus about 1% price impact)
- New pool price: $(5\,000\,000 - \Delta y)/2\,020 = 2450.8$, i.e. -1.97%
 - ▶ An arbitrageur now buys Ether in the pool and sells it elsewhere until p is back inside the no-arbitrage band that the fee sets around the market price (Seminar 16, A3)

Slippage and Impermanent Loss

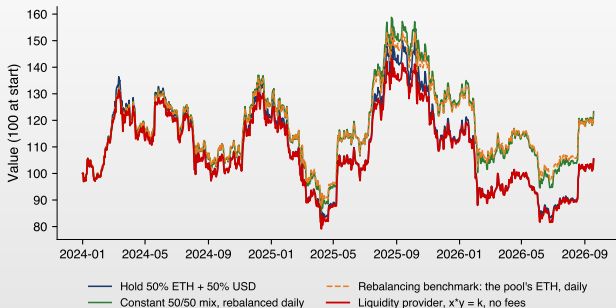


- Left: shortfall by trade size and fee; with a 0.3% fee 0.40% for 0.1% of the reserve, 1.28% for 1%, 9.34% for 10% (0.05% fee: 1.04% at 1%)
- Small trades pay the fee, large ones the price impact (about δ): liquidity is the depth of the reserve, not an order book (Chapter 10); concentrated liquidity deepens the pool near the price ([Uniswap v3](#))
- Right: value of a liquidity position relative to holding the two tokens (next slide)

Impermanent Loss and Loss Versus Rebalancing

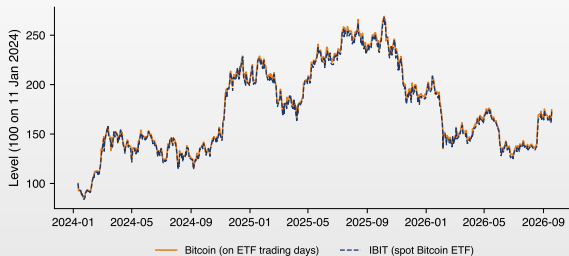
- A **liquidity provider** (LP) deposits x_0 and $y_0 = p_0 x_0$ (half and half in value) and receives a share of the pool
 - ▶ Without fees, arbitrage moves the pool to the market price p_1 : from $xy = k$ and $y/x = p_1$, $x_1 = \sqrt{k/p_1}$, $y_1 = \sqrt{k p_1}$
- LP value: $V_{LP} = p_1 x_1 + y_1 = 2\sqrt{k p_1}$; holding the tokens: $V_H = p_1 x_0 + y_0$
- With $r = p_1/p_0$ and $k = p_0 x_0^2$: **impermanent loss** $IL(r) = \frac{V_{LP}}{V_H} - 1 = \frac{2\sqrt{r}}{1+r} - 1 \leq 0$
 - ▶ $r = 2$ or $r = 0.5$: -5.72%; $r = 4$ or 0.25: -20.00%; symmetric in $\ln r$
- “Impermanent”: it vanishes if the price returns to p_0 ; it becomes permanent when the LP withdraws
- A fairer benchmark than holding: a **rebalancing** portfolio that holds the pool's quantity of Ether at every instant and trades at market prices [Milionis et al. \(2022\)](#)
 - ▶ The pool rebalances too, but at stale prices set by arbitrageurs: it always trades at a loss to them
- For $x \cdot y = k$ and a price with volatility σ , the loss versus rebalancing (LVR) grows at the rate $LVR/V = \sigma^2/8$ per unit of time [Milionis et al. \(2022\)](#)
- Fees must exceed $\sigma^2/8$ a year for a liquidity provider to beat this benchmark (not to earn a positive return): a volatility-selling business
- Derivation by Itô in the [Appendix](#); a weighted pool $x^w y^{1-w} = k$ loses at the rate $\sigma^2 w(1-w)/2$ (Seminar 16, A6)

A Liquidity Provider in an Ether Pool



- Daily Ether prices from 1 January 2024; 100 invested half in Ether, half in USD; pool with no fees; dashed: the LVR benchmark (the pool's Ether, rebalanced daily)
- Ether 2352 $\rightarrow$ 2611 USD ($r = 1.11$): impermanent loss only -0.14%; final values: holding 105.5, pool 105.4, LVR benchmark 122.4
- Realised LVR 5.83% a year against $\sigma^2/8 = 5.79\%$ ($\sigma = 68.1\%$); 95% block-bootstrap CI [4.88, 6.98]%
- Impermanent loss vanishes when the price returns; LVR accumulates with every move: fees must earn about 5.79% a year to match the benchmark

IBIT Follows Bitcoin



- Bitcoin and IBIT on ETF trading days, both 100 on 11 Jan 2024
- A **spot ETF** holds the coins with a custodian and trades in US hours; BlackRock's IBIT charges a sponsor fee of 0.25% a year [BlackRock \(2026\)](#)
- **Authorised participants** create or redeem shares against coins, which keeps the price within a cost band around the net asset value, as for a stablecoin

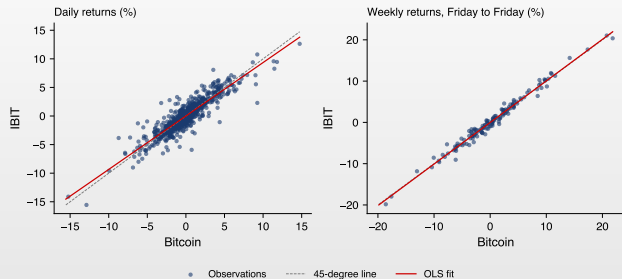
Measuring How Well an ETF Tracks

- Regression on the ETF's trading days: $r_t^{\text{ETF}} = a + \beta r_t^{\text{coin}} + u_t$; a perfect tracker has $\beta = 1$, $R^2 = 1$
- Tracking error** $\text{TE} = \text{sd}(r_t^{\text{ETF}} - r_t^{\text{coin}})\sqrt{m}$ ($m = 252$ daily, 52 weekly); **fee drift**: slope of $\ln(P_t^{\text{ETF}}/P_t^{\text{coin}})$ on time in years (the fund sells coins to pay its fee)

	$\hat{\beta}$ daily	R^2 daily	$\hat{\beta}$ weekly	R^2 weekly	TE daily	TE weekly
IBIT / Bitcoin	0.93	0.82	1.01	0.98	20.8	6.2
ETHA / Ether	0.94	0.83	0.99	0.98	29.9	9.0

- Weekly data show a near-perfect tracker; fee drift -0.24% a year (IBIT) and -0.24% (ETHA), close to the 0.25% fee
- IBIT trades 2.2 billion USD a day on average (maximum 10.3 billion on 5 Feb 2026)

Daily vs Weekly Returns and the Dimson Beta



▣ IBIT against Bitcoin: daily returns on common days (left), Friday-to-Friday returns (right); R^2 0.82 vs 0.98

▣ The coin's daily close and the New York close are hours apart: part of day t 's move reaches the ETF on $t \pm 1$, which biases the daily OLS beta towards zero; Newey–West fixes the SE, not the bias

▣ Dimson (1979):

$$r_t^{\text{ETF}} = a + \sum_{j=-1}^1 b_j r_{t+j}^{\text{coin}} + u_t, \text{ sum beta}$$

$$\hat{\beta}_D = \hat{b}_{-1} + \hat{b}_0 + \hat{b}_1$$

▣ IBIT: $\hat{\beta}_D = 1.00$ (SE 0.023) vs OLS 0.94; ETHA: 0.99 (SE 0.022) vs 0.94: within one SE of 1, as the weekly betas; where prices form is a separate question Hasbrouck (1995)

 MFM_ch16_etf

Crypto Equities: Coinbase and Strategy

- ▣ **Coinbase**: a US crypto exchange, revenue tied to trading volume; **Strategy** (formerly MicroStrategy): a software company holding Bitcoin as its main treasury asset

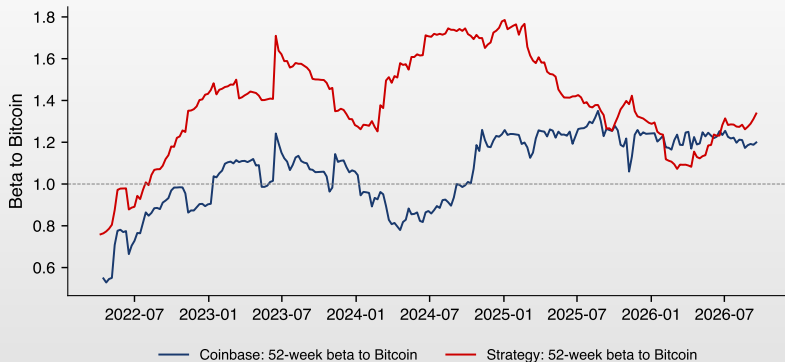
- ▣ Weekly two-factor regression, Newey–West SE, April 2021 – September 2026:

$$r_t^{\text{stock}} = a + \beta_{\text{BTC}} r_t^{\text{BTC}} + \beta_{\text{SPX}} r_t^{\text{S\&P}} + u_t$$

	$\hat{\beta}_{\text{BTC}}$	SE	$\hat{\beta}_{\text{SPX}}$	SE	R^2	σ_a (%)
Coinbase	0.79	0.09	2.05	0.22	0.52	84
Strategy	1.18	0.11	1.02	0.24	0.63	89

- ▣ Strategy: 1.18 times Bitcoin plus 1.02 times the S&P 500, a leveraged Bitcoin position with extra equity risk
- ▣ Coinbase: $\hat{\beta}_{\text{SPX}} = 2.05$, a high-beta US stock whose business depends on crypto activity, more an equity than a Bitcoin proxy

Beta to Bitcoin Over Time



- Rolling 52-week beta of weekly returns on Bitcoin; Strategy's ranged from 0.76 to 1.79; both stocks are more volatile than Bitcoin (55% a year on the same weeks)
- When did the beta change? A break test at an unknown date: Seminar 16, B7–B8

Testing a Factor Model: GRS and Fama–MacBeth

- N test assets, K traded factors, T periods: $r_{it}^e = \alpha_i + \beta_i' f_t + \varepsilon_{it}$; model adequacy $H_0: \alpha_1 = \dots = \alpha_N = 0$
 - ▶ $GRS = \frac{T - N - K}{N} \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{f}' \hat{\Omega}^{-1} \bar{f}} \sim F(N, T - N - K)$ [Gibbons, Ross & Shanken \(1989\)](#); $\hat{\Sigma}$, $\hat{\Omega}$: residual and factor covariances
 - ▶ F is exact only for i.i.d. errors with the Normal distribution: with crypto tails, add a wild bootstrap under H_0
- Premium of a factor: Fama–MacBeth, two passes; $\hat{\lambda}$ = mean of the weekly cross-sectional slopes on $\hat{\beta}_i$
 - ▶ The FM standard error ignores that $\hat{\beta}_i$ is estimated: [Shanken \(1992\)](#) correction; weak factors (small spread of β) over-reject [Kleibergen \(2009\)](#)
- GRS judges the model, FM the premium: a priced factor in an incomplete model can fail GRS; formulas and robust versions in the [Appendix](#)

Worked Result: Eight Coins and the Crypto Market Factor

- Test assets: Bitcoin, Ethereum, XRP, BNB, Cardano, Dogecoin, Litecoin, Chainlink; weekly excess returns over the 4-week bill ([FRED DTB4WK](#)), $T = 450$; factor: the five-coin index of Section 3
 - ▶ CSMB and CMOM of [Liu, Tsyvinski & Wu \(2022\)](#) need long–short portfolios over a broad cross-section with circulating market values: not buildable here
- GRS = 2.16: $p = 0.029$ from $F(8, 450 - 9)$, $p = 0.018$ from the wild bootstrap: zero alphas rejected at 5%
 - ▶ Largest alphas: Dogecoin +99% a year ($t = 1.23$), BNB +51% ($t = 1.97$); no single $|t|$ above 1.97
- Fama–MacBeth: $\hat{\lambda} = 163\%$ a year, SE 147 (FM) and 156 (Shanken); the factor itself earned 39% ($t = 1.81$)
 - ▶ Betas range only from 0.90 to 1.40: eight coins cannot identify a premium
- Survivorship (coins chosen because they are large today) may contribute to the rejection; GRS does not identify its cause

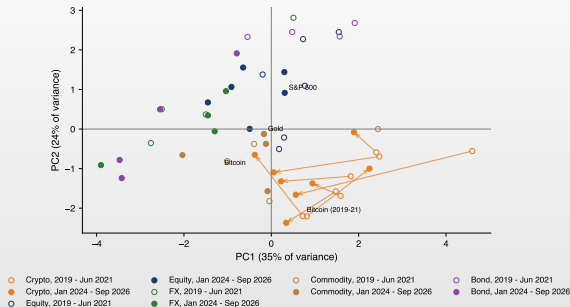
A Separate Asset Class? The Evidence So Far

- [Pele et al. \(2023\)](#) compare cryptos with stocks, exchange rates, commodities, bonds and real-estate indices
 - ▶ Dimension reduction: most of the variation is explained by **tail**, **memory** and **moment** factors of log returns
 - ▶ Classification methods place cryptos in a separate class, mainly because of the tail factor
 - ▶ The tail factor is a feature-space factor (76% of the variance), loading on quantiles, conditional tail expectations, variance and the α -stable tail and scale parameters ([Pele et al. \(2023\)](#), Sect. 3.2): it mixes scale with shape and is not a difference in Hill indices
 - ▶ Over time cryptos became more alike and more distinct from other assets
- Hedge or safe haven? A hedge is uncorrelated or negatively correlated on average; a safe haven on the worst market days [Baur & Lucey \(2010\)](#)
- Early evidence: Bitcoin a diversifier, a hedge only in some periods [Bouri et al. \(2017\)](#); tested with post-ETF data in Seminar 16, C1

A Small Replication: Features and Method

- ▣ 26 assets: 8 cryptos, 6 equity indices (including BET), 4 exchange rates, 4 commodities, 4 bond ETFs; two windows
 - ▶ W1: January 2019 – June 2021; W2: 11 January 2024 – 18 September 2026 (after the spot ETFs)
- ▣ Eight features of daily log returns, each asset on its own calendar
 - ▶ Moments: $\ln \sigma_a$, skewness, $\ln(1 + K)$; memory: $\hat{\rho}_1(r)$, $\hat{\rho}_1(r^2)$
 - ▶ Tails: Hill indices of losses and gains (5% largest); maximum drawdown
- ▣ **Principal component analysis** (PCA) of the standardised features (Chapter 3): directions of maximum variance
 - ▶ Distance between class centroids in the full 8-dimensional standardised space

Cryptos Among Other Asset Classes



- First two principal components (35% and 24% of the variance); hollow: W1 (2019 – June 2021), filled: W2 (January 2024 – September 2026); arrows: each crypto
- PC1 measures volatility, kurtosis, drawdown depth and tail heaviness; cryptos sit apart in both windows, as in [Pele et al. \(2023\)](#)
- Point estimates move towards the others: distance to equities $3.03 \rightarrow 2.63$, to commodities $2.26 \rightarrow 1.86$; Bitcoin's volatility $77\% \rightarrow 47\%$
- Cryptos became more alike (spread around their centre $2.13 \rightarrow 1.43$); median Hill index $2.32 \rightarrow 3.26$: larger than sampling noise? Next slide

 MFM_ch16_alt_assets

Is the Move Significant? A Block Bootstrap of the Map

- In each window, draw the same calendar days for all 26 assets (moving blocks of 28 days: four weeks, 20 trading days); each asset keeps its own returns on those days, so cross-asset dependence is preserved; recompute the eight features, the common standardisation and the centroid distances; 500 replications
 - ▶ Statistic: change $\Delta = d_{W2} - d_{W1}$ in the distance between the crypto centroid and each class centroid
- Equities: $\Delta = -0.39$, 95% CI $[-1.04, 0.59]$; commodities: -0.41 $[-1.38, 0.47]$; both contain 0
 - ▶ Currencies: -0.89 $[-1.67, 0.18]$; bonds: -0.62 $[-1.55, 0.92]$
 - ▶ Spread of the cryptos around their centre: -0.70 $[-1.09, 0.31]$
- Reading: all five indicative intervals contain 0; the map shows cryptos apart, but it does not establish a reliable post-ETF move towards the other classes
- Caveat: $\ln(1 + K)$ and $\hat{\rho}_1(r^2)$ have population targets only if $Er^4 < \infty$, doubtful here: they enter as sample descriptors, so the map is descriptive and these intervals are indicative

Case Study: Three Factors for Crypto Returns

- Liu, Tsyvinski & Wu (2022), *Journal of Finance*: a landmark study that applies the equity factor-model toolkit (portfolio sorts, factor regressions, GRS) to the whole cross-section of coins
 - ▶ Question: do a few crypto-specific factors price the expected returns of all coins, as market, size and momentum do for stocks?
- It builds on Liu & Tsyvinski (2021): aggregate crypto returns barely load on stock, currency or metal factors, but are predicted by crypto momentum and attention
 - ▶ The step: from the time series of three coins to the cross-section of 1,827 coins
- Why it matters: a factor model is the benchmark for every claimed crypto anomaly and for judging crypto funds
 - ▶ If coins are segmented from equities, stock factors are the wrong benchmark
- Link to this chapter: the market-value index of Section 3; CMKT, CSMB, CMOM, the GRS test of Section 7; data-quality threats in the [Appendix](#)

Case Study: Data and Method

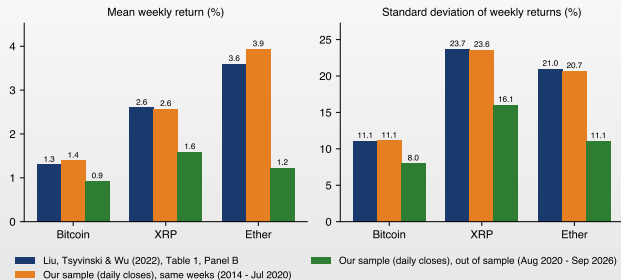
- Data: daily prices, volumes and market values of active and defunct coins from an aggregator of over 200 exchanges (Liu, Tsyvinski & Wu (2022), Sect. 2); market value of at least 1 million USD: 1,827 coins, first week of 2014 – July 2020
 - ▶ Weekly returns from daily closes: 52 weeks a year, the last one of eight days (nine in leap years); CMKT: value-weighted return of all coins minus the one-month bill rate
- Anomalies: 24 characteristics from the equity literature that need only price, volume and market value (Table 2); each week, value-weighted quintile portfolios and a top-minus-bottom spread
 - ▶ CSMB: bottom 30% minus top 30% of coins by market value; $CMOM = \frac{1}{2}(SH + BH) - \frac{1}{2}(SL + BL)$ from 2×3 sorts on size and past three-week return (S/B: small/big coins; H/L: high/low past three-week return; Liu, Tsyvinski & Wu (2022), Sect. 4.1, Eq. 1)
- Tests: the ten significant spreads regressed on one-, two- and three-factor models (Tables 8–9); joint GRS tests on all 24 spreads (Table 11)
 - ▶ Multiple testing: p -values adjusted under a generalised FWER (k -FWER, probability of k or more false rejections, Lehmann & Romano (2005)), weaker than Holm's FWER

Case Study: Key Results

Result	Numbers in the paper	Reading
Weekly returns (Table 1)	mean, s.d.: market 1.3%, 11.2%; Bitcoin 1.3%, 11.1%; XRP 2.6%, 23.7%; Ether 3.6%, 21.0%	the market behaves like Bitcoin
Size (Table 3)	top minus bottom quintile -5.8% a week ($t = -2.45$)	premium in the smallest coins
Momentum (Table 4)	three-week winners minus losers 3.1% a week ($t = 2.65$)	winners keep winning
Crypto CAPM (Table 8)	all ten alphas significant; momentum $\alpha = 3.0\%$ ($t = 2.55$), $R^2 = 0.001$	the market alone fails
Three factors (Table 9)	no alpha significant; momentum $\alpha = 1.1\%$ ($t = 1.41$), CMOM loading 1.12, $R^2 = 0.58$	size and momentum explain the ten anomalies
Joint test (Table 11)	GRS on 24 spreads: $p = 0.033$ (Fama–French six factors), $p = 0.194$ (crypto three)	stock factors are the wrong benchmark

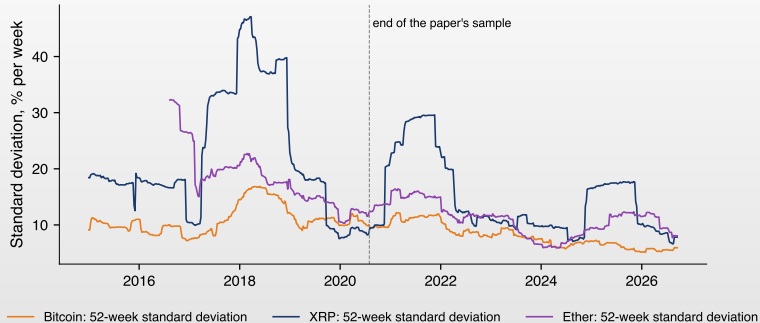
- Momentum is a large-coin effect: 3.2% a week above the median size, insignificant below (Table IA.10), the opposite of stocks

Case Study: Table 1 Rebuilt on the Course Data



- Weekly simple returns on the paper's calendar; blue: Table 1, Panel B of [Liu, Tsyvinski & Wu \(2022\)](#); orange: our sample (daily closes), same weeks; green: August 2020 – 16 Sep 2026
- Same weeks: mean Bitcoin 1.4% (paper 1.3%), XRP 2.6% (2.6%), Ether 3.9% (3.6%); s.d. 11.1, 23.6, 20.7% (paper 11.1, 23.7, 21.0)
- Largest gap, the Ether mean: 0.26 i.i.d. standard errors (SE 1.29 points); this checks the data and the calendar, not the factors
- Out of sample (319 weeks): means 0.9, 1.6, 1.2%; s.d. 8.0, 16.1, 11.1%: Ether's volatility roughly halves

Case Study: Volatility Before and After the Paper's Sample



- Standard deviation of weekly returns over the last 52 weeks; dashed: end of July 2020, the end of the paper's sample; peaks: XRP 47.1% (Mar 2018), Bitcoin 16.9% (May 2018) a week
- Last window: Bitcoin 5.9%, XRP 7.8%, Ether 8.0% a week; volatility moves in regimes: factor tests across 2020 need HAC errors and a break test at an unknown date (Seminar 16, C3)

Case Study: Replicate, Limits, Extension

- **Replicate** with the course data: daily closes of the nine non-stable coins (Bitcoin, Ether, XRP, BNB, Cardano, Solana, Dogecoin, Litecoin, Chainlink); one-month Treasury bill rate from the [Kenneth French Data Library](#)
 - ▶ Table 1, Panel B for Bitcoin, XRP and Ether: previous slides; next, the median, skewness and kurtosis, and the CMKT proxy: the five-coin market-value index of Section 3
 - ▶ Data limit: CSMB and CMOM need breakpoints over hundreds of coins with market values, dead coins included; the nearest published design is the top-20 tercile of Table IA.13, but our nine coins are today's survivors
- **Limits** stated by the authors (Section 6.2)
 - ▶ A short sample, 2014 – July 2020, while investors were still learning about the technology: only the split-half check (same signs)
 - ▶ Most coins cannot be shorted; for the top 20 coins, after trading costs, about 60% of the returns of long-short strategies that short Bitcoin remain (Table IA.16)
 - ▶ With prices from three reputable USD exchanges, the market-value size spread is no longer significant (Table IA.15)
- **Open extension:** out of sample, August 2020 – September 2026 (the 2022 collapses, the spot ETFs), does three-week momentum among large coins survive, and is CMKT still segmented from equities?
 - ▶ Crypto-CAPM alphas; the segmentation regression of footnote 16 of [Liu, Tsyvinski & Wu \(2022\)](#) on the global five factors [Fama & French \(2015\)](#) and five exchange rates; a break test at an unknown date (Seminar 16, C3)

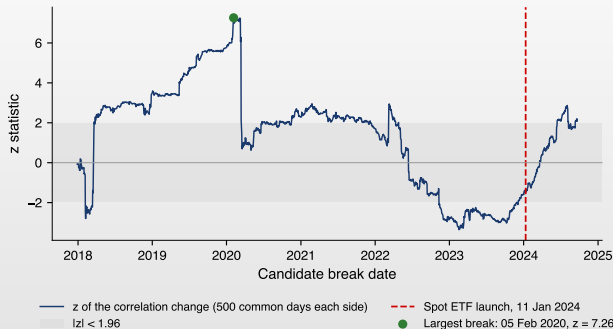
An Open Question and the Discovery Loop

- Did the US spot Bitcoin ETFs (since 11 January 2024) integrate Bitcoin with equities, in the mean, in the tails and in price discovery?
 - ▶ A live question: [Lee & Lee \(2025\)](#) study the shifted links after the approval; [Kang & Lee \(2026\)](#) title their answer “institutionalization without integration”
- Formal version: $H_0: \rho_{\text{post}} = \rho_{\text{pre}}$ against $H_1: \rho_{\text{post}} > \rho_{\text{pre}}$; the same for the lower-tail dependence λ_L of Chapter 6
 - ▶ Falsified if the data choose a break far from January 2024, or if placebo dates give changes as large
- Identification: the launch is an event, not an experiment; it overlaps with the rate cycle, the April 2024 halving and flows that respond to prices
- The discovery loop in one pass, the assistant drafts and the researcher verifies
 - ▶ literature (resolve every DOI), competing hypotheses, code that first replicates $\hat{\rho} = 0.39$ and 0.38, a pre-registered robustness plan, a referee-style critique; reusable prompts in the [Appendix](#)

What the Researcher Must Check

- ▣ **Calendars:** returns computed on each calendar and then matched pair Bitcoin's Sunday-to-Monday move with the S&P 500's Friday-to-Monday move; align prices on common trading days first, then compute returns
- ▣ **Closing times and volatility:** non-synchronous closes bias daily correlations towards zero, high volatility raises them [Forbes & Rigobon \(2002\)](#): check weekly returns, the lead-lag correlation and the adjusted ρ^*
- ▣ **Survivorship and supply:** use the universe as it was on each date and the circulating supply, as in the CRIX rule
- ▣ **Invented references and numbers:** every DOI resolves on Crossref with the same title; every number is regenerated by code in the repository

Mini-Case: Did the ETFs Break the Correlation?



- Assistant's claim: "the spot ETFs integrated Bitcoin with equities"; $\hat{\rho} = 0.39$ in 2020–2023 ($n = 1006$), 0.38 since 11 January 2024 ($n = 674$)
- Difference -0.01 , 95% CI $[-0.12, 0.13]$ (circular block bootstrap, blocks of 20 days)
- Chart: $z(\tau) = \frac{[\text{atanh } \hat{\rho}_{\text{after}} - \text{atanh } \hat{\rho}_{\text{before}}]}{\sqrt{2/(h-3)}}$, $h = 500$ days on each side
- Largest break 5 Feb 2020 ($z = 7.26$); at the ETF date $z = -1.48$ and 82% of candidate break dates give a larger $|z|$: the integration came in early 2020; tails and price discovery remain open

 ai_discovery_case.py

Your Turn: a Project Seed

- ▣ **Question:** did institutionalisation change Bitcoin's **tail** dependence and **price discovery** with US equities?
- ▣ **Replicate first:** $\hat{\rho} = 0.39$ and 0.38 from this chapter, then one headline estimate of [Lee & Lee \(2025\)](#) or [Kang & Lee \(2026\)](#) on their own sample
 - ▶ Extend: lower-tail dependence from copulas (Chapter 6), weekly returns, IBIT–Bitcoin lead–lag in US trading hours
- ▣ **Deliverables:** a pre-registered plan committed before the results, a notebook that regenerates every number, effect sizes with CIs and the full list of specifications
- ▣ **AI log:** prompts that shaped the design in `AI_USE.md`; each wrong claim (an invented DOI, returns on mismatched calendars) and the test that caught it in `AI_ERRORS.md` ([templates](#))

Key Takeaways

- Crypto returns: 3–7 times the volatility of shares, drawdowns of 80–96%, heavy tails in all six assets (Hill near 2.7), calmer weekends; the link with equities broke in March 2020, not at the ETF launch
- Indices: a Laspeyres-type index with a divisor; CRIX picks its size by an AIC on tracking residuals; value weights make crypto indices mostly Bitcoin
- Stablecoins: the peg is an arbitrage band that holds only while redemption works; a threshold AR fits it, tested with a bootstrap, not with χ^2
- DeFi: $xy = k$ prices a swap as fee plus price impact; providing liquidity is selling volatility, at $\sigma^2/8$ a year
- ETFs track one for one once closing times are handled; cryptos remain a separate class in distribution, a risk asset in co-movement

Check Yourself, and Next: Chapter 17

Retrieval questions

- Why is Bitcoin's sample kurtosis not reported, and what replaces it?
- Why can a threshold or a break date not be tested with χ^2 critical values?
- Why does a daily regression understate how well IBIT tracks Bitcoin?

Next chapters

- Chapter 17, bubbles and crashes: explosive-root tests and log-periodic power laws, crypto booms as a main case
- Chapter 18, systemic risk: spillovers between crypto, banks and equity markets
- Revisit: VaR and ES for crypto (Chapter 7, [Pele & Mazurencu-Marinescu-Pele \(2019b\)](#)), liquidity on crypto exchanges (Chapter 10, [Agakishiev, Härdle & Pele \(2025\)](#)); reading: [Härdle, Harvey & Reule \(2020\)](#), [Franke, Härdle & Hafner \(2019\)](#), [Schär \(2021\)](#)

Appendix: Tail Index, Inference Conditions

- Under i.i.d. data with a regularly varying tail, $k \rightarrow \infty$, $k/n \rightarrow 0$, and a second-order condition with negligible bias ($\sqrt{k} A(n/k) \rightarrow 0$): $\sqrt{k}(\hat{\alpha}_H - \alpha) \Rightarrow N(0, \alpha^2)$; the 5% share is a finite-sample tuning choice
 - ▶ With volatility clustering the variance depends on extremal dependence and is typically larger [Drees \(2000\)](#); a block bootstrap needs further tail and block-length conditions; choose k from a Hill plot [Drees, de Haan & Resnick \(2000\)](#)
 - ▶ Bitcoin: $\hat{\alpha}_H = 2.73$ ($k = 78$), SE 0.31; with $k = 2.5\%$ or 10% : 3.48 and 2.59: the choice of k matters as much as the sampling error
- Block bootstrap intervals: up to 4.48 for the S&P 500; bootstrap CIs for the sample kurtosis are invalid when $\alpha < 4$ (bootstrap failure with infinite variance, [Athreya \(1987\)](#))
 - ▶ Dogecoin: sample kurtosis 106.4, driven by a few social-media jumps, with $\hat{\alpha}_H = 2.83$: it estimates nothing stable
- Equal tails? Bitcoin vs S&P 500: $z = -0.28$; Dogecoin vs S&P 500: $z = -0.07$ (i.i.d. SE, independent estimates)
 - ▶ Overlapping CIs are not a test of equality, and non-rejection is not equivalence: that needs a dependence-aware interval for the difference, robust to the choice of k

Appendix: The Published CRIX Rule in Detail

- Total market index TMI (all coins) and candidate indices $\text{CRIX}(k, \beta)$: the k_1 largest coins at weight 1, the next $s = k - k_1$ with free weights β (Trimborn & Härdle (2018), Sect. 3, eqs. (11)–(12) in Trimborn & Härdle (2018, author version))
 - ▶ $\hat{\varepsilon}_t(k, \beta) = \Delta \ln \text{TMI}_t - \Delta \ln \text{CRIX}_t(k, \beta)$; $\hat{\beta}$ minimises $\|\hat{\varepsilon}(k, \beta)\|^2$
- $\text{AIC}(k) = -2 \log L\{\hat{\varepsilon}(k, \hat{\beta})\} + 2s$ (eq. (20)); L uses an Epanechnikov kernel density of $\hat{\varepsilon}(k_1, 1)$ with a Sheather–Jones plug-in bandwidth
 - ▶ Under Normal errors it reduces to $T \ln \hat{\sigma}_k^2 + 2s$ (Trimborn & Härdle (2018, author version), Appendix, eq. (35))
- ECRIX: $k_1 = 1$, steps of one coin, stop at the first rise of the AIC (Trimborn & Härdle (2018, author version), Sect. 5, rule (26)); EFCRIX: global minimum (rule (27))
 - ▶ Members reallocated monthly by market value; k reviewed quarterly on the last three months of data (Trimborn & Härdle (2018), Sect. 4)
- Data limitation: circulating supply is public for five of our coins only (Bitcoin, Ethereum, Dogecoin, Cardano, Litecoin), so $k \leq 4$ and TMI has five members, not hundreds

Appendix: Peg Deviations, Summary Statistics

	s.d. (bp)	$ d_t > 50$ (%)	lowest close (bp)	lowest low (bp)
Tether	7.3	0.14	-41	-515
USD Coin	7.5	0.14	-285	-1226
Dai	10.8	0.57	-261	-1030

□ Daily close, 1 January 2021 – 18 September 2026; $d_t = 10^4(P_t - 1)$ in basis points; lowest low from the daily low price

Appendix: Tokenisation of Real-World Assets

- ▣ **Tokenised yield-bearing instruments:** claims on Treasury bills and cash issued as tokens on a blockchain; the legal form differs
 - ▶ Fund shares: BlackRock BUIDL 2.38, Circle USYC 2.55; a tokenised secured note issued by a special-purpose vehicle: Ondo USDY 2.22 ([Ondo](#)) (billion USD, 18 Sep 2026, [DefiLlama](#))
- ▣ Unlike a payment stablecoin, they pay the Treasury yield to the holder
- ▣ Benefits: settlement 24/7, use as collateral in DeFi; risks: smart-contract bugs, legal ownership, liquidity in stress [Aramonte, Huang & Schrimpf \(2021\)](#)
- ▣ Token economics and adoption: [Cong, Li & Wang \(2021\)](#)

Appendix: Decentralised Finance in Five Layers

- Schär (2021) describes DeFi as a stack of layers
 - ▶ **Settlement:** the blockchain and its native coin (Ether)
 - ▶ **Assets:** tokens issued on it (stablecoins, governance tokens)
 - ▶ **Protocols:** exchanges, lending, derivatives as smart contracts
 - ▶ **Applications and aggregators:** user interfaces that route trades across protocols
- Promise: open access, transparency, composability (protocols plug into each other)
- Risks: high leverage, liquidity mismatches, built-in interconnectedness, no shock absorber [Aramonte, Huang & Schrimpf \(2021\)](#)
 - ▶ “Decentralisation illusion”: governance tends to concentrate power [Aramonte, Huang & Schrimpf \(2021\)](#)

Appendix: Deriving $LVR = \sigma^2/8$

- Pool value at market price P after arbitrage: $V(P) = Px + y = 2\sqrt{kP}$, concave: $V'(P) = \sqrt{k/P} = x$, $V''(P) = -\frac{1}{2}\sqrt{k}P^{-3/2}$
 - ▶ Rebalancing portfolio: holds the same $x_t = V'(P_t)$ Ether at every instant, so $dR_t = V'(P_t) dP_t$
- Price $dP_t = \mu P_t dt + \sigma P_t dW_t$; Itô: $dV_t = V' dP_t + \frac{1}{2} V'' \sigma^2 P_t^2 dt$
 - ▶ $d(R_t - V_t) = -\frac{1}{2} V''(P_t) \sigma^2 P_t^2 dt = \frac{1}{4} \sqrt{kP_t} \sigma^2 dt = \frac{\sigma^2}{8} V_t dt$
- LVR is the pool's negative gamma: a short-volatility position that loses $\sigma^2/8$ of its value per unit of time, whatever the drift [Milionis et al. \(2022\)](#)
- Assumptions: continuous geometric Brownian motion, frictionless arbitrage, no fees, continuous blocks
 - ▶ With fees and discrete blocks, arbitrageurs trade only when the price leaves the fee band, which lowers the loss [Milionis, Moallemi & Roughgarden \(2025\)](#)
 - ▶ Weighted pool $x^w y^{1-w} = k$: rate $\sigma^2 w(1-w)/2$ (Seminar 16, A6)

Appendix: The Liquidity-Provider Benchmarks in Detail

- ▣ Realised LVR rate $\sum_t (\Delta R_t - \Delta V_t) / V_{t-1}$ per year; cumulative LVR 17.0 points over 2.7 years
- ▣ The match with theory is not a test: each day's loss is $R_t/2 - (\sqrt{1 + R_t} - 1) \approx R_t^2/8$, so the realised LVR is the realised variance over 8 (bootstrap correlation of the two: 0.9981)
- ▣ 2.7 years pin the realised rate down only to about ± 1 point: CI [4.88, 6.98]%
- ▣ The constant 50/50 mix (final value 123.3) is a different benchmark with the same instantaneous rate: its log growth exceeds the pool's by $\sigma^2/8$, here 5.79% a year

Appendix: Crypto Factors, the Literature

- [Liu & Tsyvinski \(2021\)](#): crypto returns have little exposure to stock, currency or commodity factors; they load on crypto momentum and investor attention
 - ▶ [Liu, Tsyvinski & Wu \(2022\)](#): three factors from weekly coin returns: CMKT (value-weighted market excess return), CSMB (small minus big coins), CMOM (past three-week winners minus losers)
- Value from on-chain adoption and segmentation from traditional markets [Cong, Karolyi, Tang & Zhao \(2026\)](#); mispricing versus risk compensation [Babiarz & Bianchi \(2025\)](#)
- Fundamentals: network value follows Metcalfe's law in the long run, not in the short run [Pele & Mazurencu-Marinescu-Pele \(2019a\)](#); adoption and token value reinforce each other [Cong, Li & Wang \(2021\)](#)
- Two different questions, tested separately on the next slides
 - ▶ Model adequacy: are all test-asset alphas zero for the chosen factors? (time series, GRS)
 - ▶ Risk premium: is a factor's premium nonzero? (a traded factor: its mean excess return; cross-section: Fama–MacBeth λ)



Appendix: Time-Series Test, GRS

- N test assets, K traded factors, T periods: $r_{it}^e = \alpha_i + \beta_i' f_t + \varepsilon_{it}$; $H_0: \alpha_1 = \dots = \alpha_N = 0$
 - ▶ $GRS = \frac{T - N - K}{N} \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{f}' \hat{\Omega}^{-1} \bar{f}} \sim F(N, T - N - K)$ [Gibbons, Ross & Shanken \(1989\)](#)
 - ▶ $\hat{\Sigma}$: residual covariance, $\hat{\Omega}$: factor covariance (both divided by T); the denominator is one plus the squared Sharpe ratio of the factors
- The F distribution is exact only if the errors, conditional on the factors, are i.i.d. with the Normal distribution; crypto returns have $\hat{\alpha}_H$ near 3 and volatility clustering, so these assumptions are doubtful for the residuals
 - ▶ Robust alternatives: a GMM Wald test with HAC covariance, or a wild bootstrap under H_0 : $r_t^{e*} = \hat{\beta}' f_t + \hat{\varepsilon}_t \eta_t$, one Rademacher η_t for all assets (keeps the cross-correlation); valid if the residuals are serially uncorrelated (martingale differences), otherwise a block or dependent wild bootstrap
- $\hat{\Sigma}$ must be invertible and precise: N much smaller than T ; weekly crypto data since 2018 give $T \approx 450$
- GRS tests the joint pricing restriction, not a factor's premium: a zero-premium factor need not cause a rejection, and a priced factor in an incomplete model can fail GRS

Appendix: Cross-Section, Fama–MacBeth, Shanken and Weak Factors

- Two passes: $\hat{\beta}_i$ from time series; then each week $r_{it}^e = \gamma_0 + \hat{\beta}_i' \lambda_t + e_{it}$; $\hat{\lambda} = \bar{\lambda}_t$, $SE_{FM} = sd(\hat{\lambda}_t)/\sqrt{T}$
 - ▶ The FM standard error ignores that $\hat{\beta}_i$ is estimated (errors in variables): too small
 - ▶ [Shanken \(1992\)](#), for $\hat{\theta} = (\hat{\gamma}_0, \hat{\lambda}')'$ and $X = [1, \hat{\beta}]$: $\widehat{Var}(\hat{\theta}) = \frac{1}{T} [(X'X)^{-1}X'\hat{\Sigma}X(X'X)^{-1}(1 + \hat{\lambda}'\hat{\Sigma}_f^{-1}\hat{\lambda}) + \hat{\Sigma}_f^*]$,
 $\hat{\Sigma}_f^* = \begin{pmatrix} 0 & 0 \\ 0 & \hat{\Sigma}_f \end{pmatrix}$; $\widehat{Var}(\hat{\lambda})$ is its lower $K \times K$ block
 - ▶ Assumptions: residuals serially uncorrelated and homoskedastic conditional on the factors, $\hat{\Sigma}$ their covariance; the residual part is multiplied by $1 + \hat{\lambda}'\hat{\Sigma}_f^{-1}\hat{\lambda}$, the factor part $\hat{\Sigma}_f^*$ is added once
- Weak or useless factors (small spread of β , e.g. attention proxies): λ is weakly identified and FM t -statistics over-reject
 - ▶ Use identification-robust tests of risk premia [Kleibergen \(2009\)](#)
- Multiple testing: dozens of crypto “anomalies” have been tried; a new factor should clear $|t| > 3$ [Harvey, Liu & Zhu \(2016\)](#), or control the false discovery rate

Appendix: Data-Quality Threats to Identification

- **Wash trading:** fake volume on unregulated exchanges inflates volume- and liquidity-based factors [Cong, Li, Tang & Yang \(2023\)](#)
 - ▶ Filter the exchanges and volumes using the rules of the cited paper before building any factor
- **Which price?** Large, recurrent gaps between exchanges, much larger across countries than within them [Makarov & Schoar \(2020\)](#)
- **Survivorship:** today's large coins are the winners; dead coins are missing from most samples [Będowska-Sójka, Wójcik & Pele \(2026\)](#)
- **Remedy:** fix the investable universe ex ante (exchange list, volume filter), keep delisted coins, report results by sub-universe

Appendix: Common Pitfalls

- Reporting sample kurtosis as a population estimate when $\hat{\alpha}_H < 4$: if $\alpha < 4$ it estimates nothing and its bootstrap CI is invalid; assess first whether $Er^4 < \infty$ is plausible
- Testing a threshold or a break date with χ^2 critical values: the nuisance parameter is not identified under H_0
- Comparing daily closes recorded at different times and concluding that an ETF tracks badly
- Reading TVL growth as adoption without separating the valuation of the tokens from net deposits
- Treating a stablecoin as risk-free cash: the peg is only as good as redemption and reserves
- Total instead of circulating supply in a market-value index; survivorship: studying only today's large coins ignores the many that failed [Będowska-Sójka, Wójcik & Pele \(2026\)](#)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature:** map what is known, then verify every reference
 - ▶ “List 2023-2026 papers that test for a change in Bitcoin-equity co-movement at the US spot ETF approval: DOI, sample, method, sign of the effect.”
 - ▶ Resolve each DOI on [Crossref](#), open the paper, find the number it is cited for; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing mechanisms, each with an implication the others do not share
 - ▶ “Give three competing mechanisms for the post-2020 Bitcoin-S&P 500 correlation and, for each, one testable implication that separates it from the others.”
 - ▶ E.g. ETF flows predict an effect concentrated in US trading hours; a common monetary factor predicts co-movement with rate surprises already before 2024
- **Data and code:** on the course data, replicating a known number first
 - ▶ “Take the daily closes of Bitcoin and the S&P 500 from the course data, align the prices on common trading days, then compute log returns and their 2020-2023 correlation.”
 - ▶ Replicate first: this chapter reports 0.39 (2020–2023) and 0.38 (since the ETFs); no new result before the code hits both



Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the post-ETF results are seen
 - ▶ “Draft a pre-registered robustness plan: break-date scan, placebo launch dates, weekly returns, tail dependence, block bootstrap; fix every tuning choice now.”
 - ▶ The plan fixes windows, block length and the family of tests: the number of specifications is known and the FWER can be controlled
- **Critique**: ask the assistant to attack the result
 - ▶ “Act as a referee: give the three strongest reasons this change in correlation could be spurious, and the test that would settle each.”
 - ▶ Expect: non-synchronous closes, correlation that rises mechanically with volatility Forbes & Rigobon (2002), confounding events
- **Report**: effect size with a CI, every specification tried, what failed
 - ▶ The prompts that shaped the design go into AI_USE.md

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Are Cryptos becoming alternative Assets?
- ▣  Introduction to Blockchain and Cryptocurrencies
- ▣  Comparison of Cryptocurrency Indices
- ▣  Cryptocurrency as an Asset Class
- ▣  Smart Contracts
- ▣  Stabilising Onchain Stablecoin Risks
- ▣  Risk Premia in the Bitcoin Market

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_16

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